The Burpee Enigma: Literature Review

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Factors affecting falling and injury during a multi-stage mountain bike event: a prospective study protocol

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measurement issues (location/orientation of the ultrasound transducer), individual issues (changes in tissue properties related to activity), or task issues (differences in task between sessions). The results highlight that care should be taken when comparing shear elastic modulus of the ITB between sessions and between participants.

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The Burpee Enigma: Literature Review

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Background: The burpee, is one of the best-known exercises worldwide due to increased popularity among high intensity interval training programs. However, previously conducted research on the burpee has used inconsistent terminology and discrepancies in technical knowledge that has resulted in uncertain data representation. This literature review is the first to collate evidence of different forms of burpees from previous research, and critically analyse the physiological and performance effects of each different version.

Methods: The preliminary search was performed electronically using Google Scholar, PubMed and Web of Science. The search was conducted on all language written papers pending legible translation, published in peer reviewed journals from January 1939-September 2018 (when the initial search started). Studies that investigated physiological responses (heart rate, blood lactate and oxygen consumption) and/or anthropometric values (height, weight, age, gender) and/or the development/history of the nomenclature of the burpee were included.

Results: From the initial 2700 results, only 19 papers were examined in depth having met the appropriate inclusion criteria.

Discussion: Synthesis of the study results indicated that when physiological responses are being measured, the type of the burpee is of great importance, because the use of different movement sequences has different metabolic demands. The traditional or the modern burpee places large metabolic and neuromuscular demands due to the compound nature of the individual movement, and utilising both upper body and lower body in unison adds to the complexity. Exercises that require greater complexity and are less familiar to the participant create more central and peripheral fatigue, that in turn affects the rate of recovery. The burpee is a high-intensity bodyweight movement that is difficult to qualify and quantify due to variation in technique, sequence, and individual athlete somatotypes. It has multiple uses across many sport specific contexts, however is often overlooked in favour of other conventional cardiovascular exercises. While previous research on the burpee has been of high quality and valuable to the strength and conditioning community, the burpee has not received appropriate attention over the years. Applicable research regarding the burpee must have consistent naming and standardised technique to remove the inconsistent terminology and research related discrepancies.

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Factors affecting falling and injury during a multi-stage mountain bike event: a prospective study protocol

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Background: Cycling is a sport that has been experiencing an increase in popularity as a mode of transport, recreational activity and professional career. Fifty percent of recreational mountain bikers have reported at least one serious acute injury related to mountain biking. Many of the injuries are a result of falling and are related to collisions, loss of control, loss of traction and mechanical failures. Balance on a bicycle is considered a perceptual-cognitive task. In mountain biking, experienced riders use proprioception more than visual input to determine their decision making. These factors need to be considered during technical multi-stage mountain bike events where cumulative physical fatigue with cognitive fatigue may affect the decision making, reaction time or control of the rider. We aim to establish the common mechanisms for falling and any associated factors during training and while competing in a multi-stage mountain bike event.

Methods: Healthy male and female riders between 18 and 60 years who are entered into the ABSA Cape Epic (eight day stage mountain bike stage race) will be recruited in Cape Town. Riders will complete two testing sessions; at baseline 8–10 weeks prior, and at 2 weeks prior to the race. Participants will complete an informed consent form, activity questionnaire and health screening questionnaire. Reaction time will be assessed using an online programme. Mass, stature, skin-folds and leg length will be measured. Static and dynamic balance will be assessed on the bicycle using recently validated novel balance tests. Agility will be assessed on a modified Illinois agility test for cyclists. A submaximal cycle test (LSCT) will be performed to assess for predicted VO2max, Peak Power Output, 40 km time trial power and heart rate recovery. All training prior to the event will be logged in Training Peaks. Any falls or near misses and resulting injuries during the training or event will be noted by the participants. Sleep quality and quantity will be monitored during the training. During the race, riders will complete short questionnaires on their performance, recovery, sleep, nutrition and wellness.

Statistical Analysis: Descriptive data of the participants will be presented and assessed for significance. Associations between fatigue and falling will be assessed during the training and competition periods. Training and competition Injury incidence will be reported per 1000 hours. Training load will be reported as acute:chronic ratio for each of the periods, training and event and the odds ratios for injured and uninjured riders will be calculated.

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The effect of physical and cognitive fatigue on bicycle balance performance: a protocol

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Background: Cycling is a complex task that requires input from multiple bodily processing systems to maintain control and balance on the bicycle. Riders use steering, body leaning and twisting, and knee movements to oppose the lateral forces on a bicycle. Balance on and control of a bicycle are key components in preventing falls